



Problem

Given a value V , you have an infinite supply of N different coin denominations. Find the minimum number of coins required to make exactly V cents.

You must also print the coins used in an optimal solution.

If it is not possible to make the value V , print -1 .

Input Format

The first line contains an integer T , the number of test cases.

For each test case: The first line contains two integers V and N .

V — the target value. N — the number of available coin

denominations. The second line contains N integers $C[i]$, representing the coin denominations.

Constraints

$1 \leq T \leq 100$ $1 \leq V \leq 10^6$ $1 \leq N \leq 10^6$ $1 \leq C[i] \leq 10^6$

Output Format

For each test case:

Print the coins used in a minimum-coin solution, separated by spaces. In the next line, print the minimum number of coins. If it is impossible to make V , print -1 .

Line: 5 Col: 7

Sample Input 0

7 2
2 4

Sample Output 0

-1

Sample Input 1

11 4
1 5 6 9

Sample Output 1

2

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Language Python 3



```
1 # Enter your code here. Read input from STDIN.
2 V,N=list(map(int, input().split()))
3 coins=list(map(int, input().split()))
4 dp=[999999]*(V+1)
5 dp[0]=0
6 for i in range(1,V+1):
7     for coin in coins:
8         if coin<=i:
9             dp[i]=min(dp[i],dp[i-coin]+1)
10
11 if dp[V]==999999:
12     print(-1)
13 else:
14     print(dp[V])
```

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?

Test case 0

Test case 1

Compiler Message

Success